

```
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill$ gcc -Wall -Wextra skill23.c -o skill23
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill$ ./skill23
```

```
=====
                        SKILL 23 DEMO
=====

Commands:
pipeline  - Run large 4-stage pipeline
multi     - Launch multiple jobs
monitor   - Monitor resource usage
fail      - Test failure detection
performance - Measure pipeline performance
arch      - Display architecture and syscalls
report    - Generate documentation report
all       - Run complete demonstration
exit      - Exit program
```

```
skill23> pipeline
```

```
===== LARGE PIPELINE =====
Pipeline: printf | grep | tr | wc
Number of stages: 4
3
Pipeline execution time: 0.019635 seconds
Pipeline status: SUCCESS
All pipeline processes synchronized.
=====
skill23>
```

```
skill23> multi
```

```
===== MULTIPLE JOBS =====
Launching 3 jobs in parallel...
Job 1 started: PID=47223
Job 2 started: PID=47224
Job 3 started: PID=47225
Job 2 completed: PID=47224 result=959998845
Job 3 completed: PID=47225 result=959998845
Job 1 completed: PID=47223 result=959998845
Jobs completed: 3/3
Total execution time: 0.041482 seconds
Multiple-job stability: PASS
=====
skill23> |
```

```
skill23> monitor
```

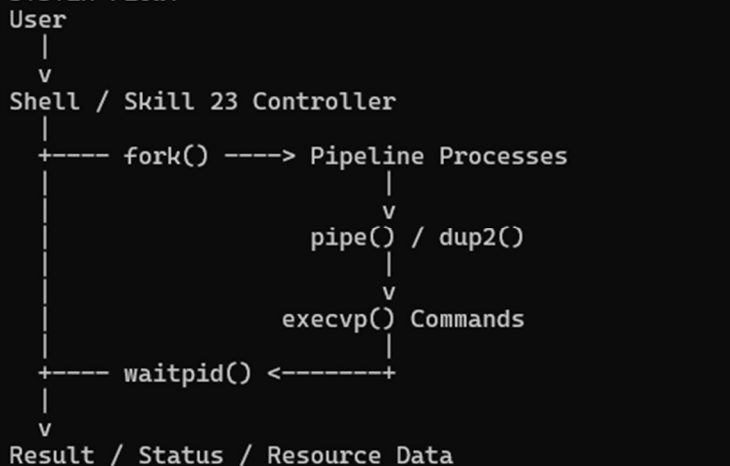
```
===== RESOURCE MONITOR =====
User CPU time   : 0.003096 sec
System CPU time : 0.000000 sec
Maximum RSS    : 2588 KB
Page faults    : 0
Voluntary ctx  : 9
Involuntary ctx : 0
Resource monitoring: PASS
=====
skill23> |
```

```
skill23> fail
```

```
===== FAILURE TEST =====  
Launching intentionally invalid command...  
exec: Permission denied  
Failure detected correctly.  
Exit status: 127  
Failure handling: PASS  
=====
```

===== ARCHITECTURE =====

SYSTEM FLOW:



PIPELINE EXECUTION PATH:

```
printf -> pipe -> grep -> pipe -> tr -> pipe -> wc
```

IMPORTANT SYSTEM CALLS:

```
fork() - create child processes  
pipe() - create IPC channels  
dup2() - connect standard streams  
execvp()- execute commands  
close() - release file descriptors  
waitpid()- synchronize processes  
getrusage()- collect resource statistics
```

DESIGN DECISIONS:

1. Each pipeline stage runs as a separate process.
2. Pipes provide ordered data flow.
3. Parent closes unused descriptors.
4. Parent waits for all children.
5. Exit statuses are checked for failures.
6. Resource usage is measured after execution.

```
Architecture review: COMPLETE
```

```
skill23> report
```

```
Report generated: skill23_report.txt  
skill23> |
```